Def. Let X be a Metric Space, Y be a Normed Space.
A sequence of functions $\{f_n : X \to Y\}$ is called pointwise bounded on X if:

for any $x \in X$, $\{f_n(x)\}$ is bounded.

And it is called uniformly bounded if: for some $M>0$,

$$\sup_X |f_n| < M \quad \text{for all } n \in \mathbb{N}.$$

Thm. If $\{f_n\}$ is a pointwise bounded sequence of complex-valued functions on a countable set E, then $\{f_n\}$ has a pointwise convergent subsequence.

Proof. Denote $E = \{x_i \mid i \in \mathbb{N}\}$. Since $\{f_n(x_1)\}$ is bounded, it has a convergent subsequence, denote by $\{f_{1,k}\}$. Similarly, $\{f_{1,k}(x_2)\}$ has a convergent subsequence, denote by $\{f_{2,k}\}$. Inductively, we obtain

$$\begin{array}{llllll} S_1: & f_{1,1} & f_{1,2} & f_{1,3} & \cdots & \text{Converges at } x_1 \\ S_2: & f_{2,1} & f_{2,2} & f_{2,3} & \cdots & \quad " \quad x_1, x_2 \\ S_3: & f_{3,1} & f_{3,2} & f_{3,3} & \cdots & \quad " \quad x_1, x_2, x_3 \\ \vdots & & & & \ddots & \quad\quad \vdots \end{array}$$

And, the order of each functions is preserved regardless of they are contained any column, by the definition of subsequence. Then the subsequence

$$S = \{f_{n,n}\}$$

Converges for each x_i. ∎

Note: $\mathscr{C}(X) = \{f : X \to \mathbb{C} \mid f \text{ continuous, bounded}\}$ where X is Metric Space.

Lemma. If K is Compact Metric Space, $f_n \in \mathscr{C}(X)$, $n \in \mathbb{N}$, and $f_n \rightrightarrows f$ on K, then $\{f_n\}$ is equicontinuous on K.

Proof. Using 3ε-method. $f_n \rightrightarrows f$ → Conti on Compact.

「Arzelà - Ascoli Theorem.

Let X be a Compact Metric space.

If $\{f_n\}$ is pointwise bounded and equicontinuous on X,

then $\{f_n\}$ is uniformly bounded and $\{f_n\}$ contains a uniformly convergent subsequence.

Proof. Given $\varepsilon>0$, there exists $\delta>0$ s.t. $d(x,y)<\delta \Rightarrow |f_n(x)-f_n(y)|<\varepsilon$ for all $n\in\mathbb{N}$.

Since X is compact, there are finitely many points $p_1,\cdots,p_r\in X$ s.t. $X=\bigcup_{i=1}^{r} B_\delta(p_i)$.

Since $\{f_n\}$ is pointwise bounded, there exist $M_i>0$ s.t. for all $n\in\mathbb{N}$, $|f_n(p_i)|<M_i$

Now, for any $x\in X$, and $n\in\mathbb{N}$, $|f_n(x)|\le|f_n(x)-f_n(p_i)|+|f_n(p_i)|\le\varepsilon+\max\{M_i \mid i=1,\cdots,r\}$.

for some p_i s.t. $x\in B_\delta(p_i)$

To show that the second assertion, Compact + Metric $\Rightarrow$ 2nd-Countable

$\Rightarrow$ Separable.

Let $E\subseteq X$ be a countable dense set. By the above Thm,

$\{f_n\}$ has a pointwise convergent subsequence on E, denote this by $\{f_{n_i}\}$. Simply, $g_i=f_{n_i}$.

Given $\varepsilon>0$, take $\delta>0$ s.t. $d(x,y)<\delta\Rightarrow|f_n(x)-f_n(y)|<\varepsilon$ for all $n\in\mathbb{N}$.

Since X is compact and E is dense in K, the set

$$\{B_\delta(x) \mid x\in E\} \text{ covers } X, \text{ and}$$

has a finite subcover, $\{B_\delta(x_i) \mid i=1,\cdots,m\}$. Since $\{g_i(x)\}$ converges for all $x\in X$,

there exists $N\in\mathbb{N}$ s.t. $i,j\ge N$, $1\le s\le m\Rightarrow|g_i(x_s)-g_j(x_s)|<\varepsilon$.

And, for any $x\in X$, for some $1\le s\le m$, $x\in B_\delta(x_s)$, thus $|g_i(x)-g_i(x_s)|<\varepsilon$ for all $i\in\mathbb{N}$.

Now, for all $x\in X$ and $i,j\ge N$,

$$|g_i(x)-g_j(x)|\le|g_i(x)-g_i(x_s)|+|g_i(x_s)-g_j(x_s)|+|g_j(x_s)-g_j(x)|<3\varepsilon. \quad \blacksquare$$